

Exercises Set: Differential Equations

Solutions

DSBA Mathematics Refresher 2026

1 First Order Differential Equations

Basics

Exercise 0: $\frac{dy}{dx} = 3x^2 - 2x + 1$.

Integrating both sides with respect to x :

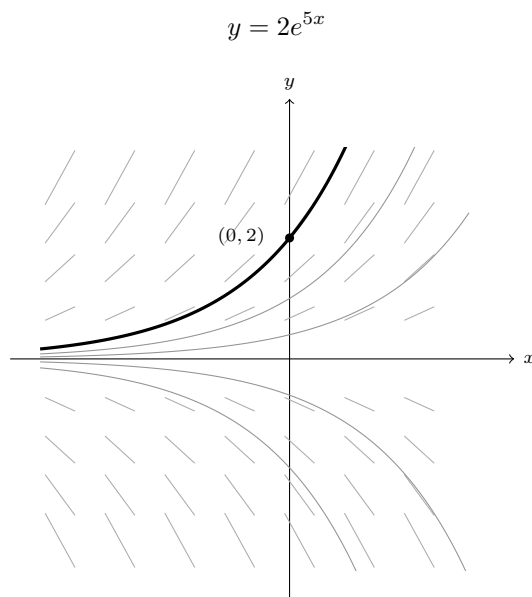
$$y = x^3 - x^2 + x + C, \quad C \in \mathbb{R}$$

Exercise 1: $\frac{dy}{dx} = 5y$.

Try $y = \alpha e^{\beta x}$; then $y' = \alpha \beta e^{\beta x} = \beta y$, so we need $\beta = 5$:

$$y = \alpha e^{5x}, \quad \alpha \in \mathbb{R}$$

With the condition $y(0) = 2$: $\alpha e^0 = \alpha = 2$, hence



The short grey segments show the slope $5y$ imposed by the equation at each point — the **direction field**. Every solution must follow it, and the grey curves are four members of the family $y = \alpha e^{5x}$. They never cross: through each point of the plane passes exactly one of them. The initial condition $y(0) = 2$ simply picks out the one through the marked point (in black).

Separable

Exercise 2: $\frac{dy}{dx} = \frac{x}{y}$.

Separating the variables and integrating:

$$\begin{aligned}y \, dy &= x \, dx \\ \int y \, dy &= \int x \, dx \\ \frac{1}{2}y^2 &= \frac{1}{2}x^2 + \frac{C}{2}, \quad C \in \mathbb{R} \\ y^2 &= x^2 + C\end{aligned}$$

so $y = \pm\sqrt{x^2 + C}$ (on any interval where $x^2 + C > 0$).

$$\text{Check: } y' = \pm \frac{2x}{2\sqrt{x^2 + C}} = \pm \frac{x}{\sqrt{x^2 + C}} = \frac{x}{y}. \quad \checkmark$$

Exercise 3: $\frac{dy}{dx} = 2x^2 e^y$.

Separating the variables and integrating:

$$\begin{aligned}e^{-y} \, dy &= 2x^2 \, dx \\ \int e^{-y} \, dy &= \int 2x^2 \, dx \\ -e^{-y} &= \frac{2}{3}x^3 - C, \quad C \in \mathbb{R} \\ e^{-y} &= C - \frac{2}{3}x^3\end{aligned}$$

so, taking logarithms,

$$y = -\ln\left(C - \frac{2}{3}x^3\right)$$

valid on any interval where $C - \frac{2}{3}x^3 > 0$.

Integrating Factor

Exercise 4: $y' + 2y = 4x$.

The integrating factor is $\text{IF} = e^{\int 2 \, dx} = e^{2x}$. Multiplying the equation by it:

$$\begin{aligned}y' e^{2x} + 2y e^{2x} &= 4x e^{2x} \\ (y e^{2x})' &= 4x e^{2x} \\ y e^{2x} &= \int 4x e^{2x} \, dx = 2x e^{2x} - e^{2x} + C, \quad C \in \mathbb{R}\end{aligned}$$

(the integral is done by parts), so

$$y = 2x - 1 + C e^{-2x}$$

$$\text{Check: } y' = 2 - 2C e^{-2x}, \text{ hence } y' + 2y = 2 - 2C e^{-2x} + 4x - 2 + 2C e^{-2x} = 4x. \quad \checkmark$$

Exercise 5: (*) $y' - \frac{1}{x}y = x^3$.

Working on $x > 0$, the integrating factor is $\text{IF} = e^{\int -\frac{1}{x} \, dx} = e^{-\ln(x)} = \frac{1}{x}$. Multiplying by it:

$$\begin{aligned}\frac{1}{x}y' - \frac{1}{x^2}y &= x^3 \cdot \frac{1}{x} = x^2 \\ \left(\frac{y}{x}\right)' &= x^2 \\ \frac{y}{x} &= \frac{1}{3}x^3 + C, \quad C \in \mathbb{R}\end{aligned}$$

so

$$y = \frac{1}{3}x^4 + Cx$$

$$\text{Check: } y' = \frac{4}{3}x^3 + C, \text{ hence } y' - \frac{1}{x}y = \frac{4}{3}x^3 + C - \frac{1}{3}x^3 - C = x^3. \quad \checkmark$$

2 Second Order Differential Equations

Basics

Exercise 0: $y'' = e^x + 4 \sin(2x) - 5x$.

Integrating once:

$$y' = e^x - 2 \cos(2x) - \frac{5}{2}x^2 + C, \quad C \in \mathbb{R}$$

Integrating a second time:

$$y = e^x - \sin(2x) - \frac{5}{6}x^3 + Cx + D, \quad C, D \in \mathbb{R}$$

Exercise 1: $y'' + 2y' + y = 0$.

Assume $y = e^{rx}$; then $r^2 e^{rx} + 2r e^{rx} + e^{rx} = 0$, i.e.

$$r^2 + 2r + 1 = 0 \iff (r + 1)^2 = 0 \iff r = -1 \text{ (double root)}$$

So $y = Ae^{-x}$ is a solution. Since the root is double, we also try $y = xe^{rx}$:

$$y' = rxe^{rx} + e^{rx}, \quad y'' = r^2 xe^{rx} + 2re^{rx}$$

$$y'' + 2y' + y = xe^{rx}(r^2 + 2r + 1) + e^{rx}(2r + 2) = 0$$

which again holds for $r = -1$. Hence $y = Bxe^{-x}$ is a solution too, and in general

$$y = Ae^{-x} + Bxe^{-x} = (A + Bx)e^{-x}, \quad A, B \in \mathbb{R}$$

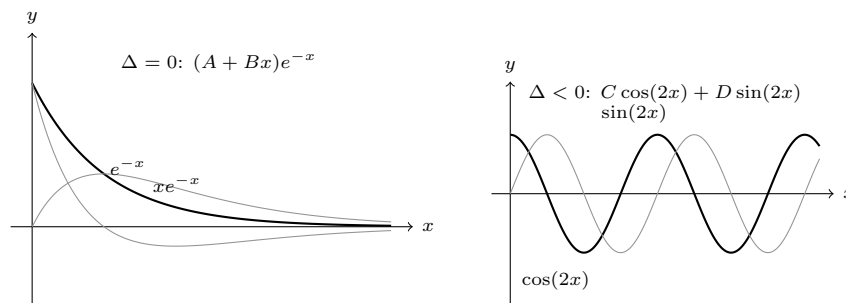
Exercise 2: $y'' + 4y = 0$.

With $y = e^{rx}$ we get $r^2 + 4 = 0$, so $r = \pm 2i$ and

$$\begin{aligned} y &= Ae^{2ix} + Be^{-2ix} \\ &= A(\cos(2x) + i\sin(2x)) + B(\cos(2x) - i\sin(2x)) \end{aligned}$$

Letting $C = A + B$ and $D = (A - B)i$, the real solutions are

$$y = C \cos(2x) + D \sin(2x), \quad C, D \in \mathbb{R}$$



The sign of the discriminant is visible in the shape of the solutions: a real root gives pure *decay* (left, with the extra factor x allowing one bump before it dies out), while a pair of complex roots gives *oscillation* (right). Here $\alpha = 0$, so the oscillation neither grows nor decays; a non-zero real part would multiply these curves by $e^{\alpha x}$ and make them spiral out or die away.

Separable

Exercise 3: $y'' = (y')^2$.

Let $z = y' = \frac{dy}{dx}$. The equation becomes $z' = z^2$, which is separable:

$$\begin{aligned} \frac{dz}{z^2} &= dx \\ \int \frac{dz}{z^2} &= \int dx \\ -\frac{1}{z} &= x + C, \quad C \in \mathbb{R} \\ z &= -\frac{1}{x + C} \end{aligned}$$

Then $\frac{dy}{dx} = -\frac{1}{x+C}$, so integrating once more:

$$y = -\ln|x+C| + D, \quad C, D \in \mathbb{R}$$

Check: $y' = -\frac{1}{x+C}$ and $y'' = \frac{1}{(x+C)^2} = (y')^2$. ✓

(Note that the constant functions $y = D$ are solutions as well.)

Non-homogeneous

Exercise 4: $y'' + 4y' + 4y = 6x^2 + 10x + 2$.

Homogeneous equation: $r^2 + 4r + 4 = 0$, i.e. $(r+2)^2 = 0$, so $r = -2$ is a double root and

$$y_{\text{hom}} = Ae^{-2x} + Bxe^{-2x}, \quad A, B \in \mathbb{R}$$

Particular solution: the right-hand side is a degree-2 polynomial, so we look for $y = ax^2 + bx + c$. Then $y' = 2ax + b$ and $y'' = 2a$, so

$$y'' + 4y' + 4y = 4ax^2 + (8a + 4b)x + (2a + 4b + 4c)$$

Identifying coefficients with $6x^2 + 10x + 2$:

$$\begin{cases} 4a = 6 \\ 8a + 4b = 10 \\ 2a + 4b + 4c = 2 \end{cases} \implies a = \frac{3}{2}, \quad b = -\frac{1}{2}, \quad c = \frac{1}{4}$$

so $y_{\text{part}} = \frac{3}{2}x^2 - \frac{1}{2}x + \frac{1}{4}$.

General solution:

$$y = y_{\text{hom}} + y_{\text{part}} = Ae^{-2x} + Bxe^{-2x} + \frac{3}{2}x^2 - \frac{1}{2}x + \frac{1}{4}$$

Exercise 5: $y'' - 2y' + 5y = 3e^{2t}$.

Homogeneous equation: $r^2 - 2r + 5 = 0$, with $\Delta = 4 - 20 = -16$, so $r = 1 \pm 2i$ and

$$y_{\text{hom}} = Ae^{(1+2i)t} + Be^{(1-2i)t} = e^t (C \cos(2t) + D \sin(2t)), \quad C, D \in \mathbb{R}$$

Particular solution: try $y = ae^{2t}$; then

$$y'' - 2y' + 5y = e^{2t} (4a - 4a + 5a) = 5ae^{2t}$$

so $5a = 3$, i.e. $a = \frac{3}{5}$ and $y_{\text{part}} = \frac{3}{5}e^{2t}$.

General solution:

$$y = e^t (C \cos(2t) + D \sin(2t)) + \frac{3}{5}e^{2t}$$

Boundary Conditions

Exercise 6: $y'' + 2y' - 3y = 0$, with $y(0) = 0$ and $y(2) = 5$.

Homogeneous equation: $r^2 + 2r - 3 = 0$, with $\Delta = 4 + 12 = 16$, so $r = \frac{-2 \pm 4}{2}$, i.e. $r = 1$ or $r = -3$:

$$y = Ae^x + Be^{-3x}, \quad A, B \in \mathbb{R}$$

Boundary conditions:

$$y(0) = 0 \implies A + B = 0 \implies B = -A$$

$$y(2) = 5 \implies Ae^2 + Be^{-6} = A(e^2 - e^{-6}) = 5 \implies A = \frac{5}{e^2 - e^{-6}}$$

Finally,

$$y = \frac{5}{e^2 - e^{-6}} e^x - \frac{5}{e^2 - e^{-6}} e^{-3x} = \frac{5(e^x - e^{-3x})}{e^2 - e^{-6}}$$

Check: $y(0) = 0$ ✓ and $y(2) = \frac{5(e^2 - e^{-6})}{e^2 - e^{-6}} = 5$. ✓

Exercise 7: (*) $y'' + 4y = 12x$, with $y(0) = 0$ and $y'(0) = 2$.

Homogeneous equation: as in Exercise 2 above,

$$y_{\text{hom}} = C \cos(2x) + D \sin(2x), \quad C, D \in \mathbb{R}$$

Particular solution: try $y = ax + b$; then $y'' = 0$ and $y'' + 4y = 4ax + 4b$, so $4a = 12$ and $4b = 0$, giving $y_{\text{part}} = 3x$.

General solution:

$$y = C \cos(2x) + D \sin(2x) + 3x$$

Boundary conditions:

$$y(0) = 0 \implies C = 0$$

$$y'(x) = -2C \sin(2x) + 2D \cos(2x) + 3, \quad \text{so} \quad y'(0) = 2D + 3 = 2 \implies D = -\frac{1}{2}$$

Finally,

$$y = -\frac{1}{2} \sin(2x) + 3x$$

Check: $y' = -\cos(2x) + 3$ and $y'' = 2 \sin(2x)$, so $y'' + 4y = 2 \sin(2x) - 2 \sin(2x) + 12x = 12x$ ✓, $y(0) = 0$ ✓, $y'(0) = -1 + 3 = 2$. ✓